Graduation research · One-page summary of findings

Artificial Intelligence, Search Engines, and Cyberchondria: A Study of Patient Pre-Consultation Behavior in Baghdad

Design: analytical cross-sectional online survey (Google Forms), Baghdad, 17 July 2026 – 11 September 2026 · **n = 300** adults · **Tools:** Short Health Anxiety Inventory (SHAI-14, 0–42) and Cyberchondria Severity Scale short form (CSS-12, 12–60; four subscales) · **Analysis:** Spearman/Pearson and partial correlation, non-parametric tests with FDR correction, robust hierarchical and logistic regression, bootstrapped mediation and moderation.

88.7%
use AI chatbots for symptoms (search engines 56.3%)

19%
clinically relevant health anxiety (SHAI ≥ 18; 95% CI 15.0–23.8%)

r = 0.47
health anxiety ↔ cyberchondria (95% CI 0.38–0.55), p < 0.001

31.7%
never tell their doctor about online/AI searches

Key findings

1. **AI is now the main gateway.** 88.7% use AI chatbots (ChatGPT 59.3% of all), ahead of search engines (56.3%); 41.0% use AI without any search engine. Convenience (80.0%), "do I need a doctor?" (57.7%) and instant reassurance (51.0%) are the main motives.
2. **Health anxiety and cyberchondria travel together.** r = 0.47; unchanged after adjusting for all covariates (partial r = 0.48). Strongest for *excessive searching* (r = 0.46) and *post-search distress* (r = 0.45); weaker for compulsion (0.25) and reassurance seeking (0.24).
3. **Health anxiety is the dominant predictor of cyberchondria** (β = 0.48); demographics explain almost nothing (R^2 = 0.030); adding health anxiety raises R^2 from 0.133 to 0.317.
4. **What drives searching is fear, not noticing.** Fear of serious illness and interpreting sensations track cyberchondria; mere awareness of aches and sensations does not (111/168 item pairs significant after FDR).
5. **Pattern consistent with a cycle:** health anxiety → excessive searching → post-search distress (indirect effect 0.16) → reassurance-seeking visits (distress almost fully mediates; direct effect 0.01). The reverse path is also significant: compatible with mutual reinforcement, not proof of it.
6. **Who is most anxious:** younger people, those with a chronic condition, social-media searchers, and those searching for instant reassurance (AOR 3.83).

Pre-consultation behaviour

- 34.7% delay or skip care because the search reassured them; 30.0% ignore the symptom.
- 25.3% visit several doctors; 22.0% seek urgent care/ER; 21.3% self-medicate.
- Higher cyberchondria → more urgent visits (AOR 1.51/SD), less ignoring (AOR 0.65/SD).
- **Concealment paradox:** higher health anxiety → *more* hiding of searches from the doctor (AOR 1.47); higher cyberchondria → *less* (AOR 0.57).
- 34.7% say searching has complicated or reduced their trust in physicians.

AI chatbots

AI users had higher cyberchondria (B = +5.3 points, adjusted), mainly excessiveness and reassurance seeking, **but** this depends on classifying 32 inconsistent answers and on only 34 non-users. AI use did not change the strength of the anxiety–cyberchondria link.

Quality of evidence

Reliable scales (α 0.81 and 0.86); CSS-12 four-factor structure confirmed; Arabic and English responders equivalent; results robust to sensitivity analyses and replicated by an independent analysis (except the AI effect).

Implications

Clinicians: ask every patient, without judgement, what they searched or asked an AI; treat repeated searching plus distress as a marker of health anxiety. **Public health:** digital and AI health-literacy campaigns in Arabic; warn about false reassurance. **AI developers:** anxiety-aware answers that avoid escalating rare diagnoses. **Research:** longitudinal studies to test the direction of the cycle.

Limitations: cross-sectional (no causality); online convenience sample, young and 44% medical students; Arabic wording not formally back-translated; single-item behaviour measures; self-report.